Comparative Study of Machine Learning Models for Predictive Modeling of Medical Data

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Abstract

This work transparently investigates and analyzes the performance of popular machine learning methods for medical diagnosis and prognosis prediction. First, a literature review was performed to choose two high-performing methods, with those being support vector machine (SVM) with a radial basis function (RBF) kernel, and random forest. These methods were then applied in an open-source implementation to freely available medical datasets, allowing for results across multiple applications to be analyzed. We found the random forest classifier achieved higher performance metrics compared to SVM with RBF across most datasets; however, both methods equally exhibited some struggle with transferability across different medical datasets.

1 Introduction

Every day, the global healthcare system generates large amounts of medical data that is often underutilized, especially considering our modern technological landscape where data is considered a highly valuable resource for analyzing problems and determining their root cause [1]. Efforts have already begun leveraging machine learning models to assist in the diagnosis and prediction of disease prognosis. They can be especially helpful in finding trends and patterns, creating connections, and making conclusions based on large datasets that would be otherwise difficult for humans to analyze manually. The complexity of the different disease mechanisms and variances in underlying symptoms across an entire patient population presents massive challenges in developing an early diagnosis tool that can fully span this needed scope [2]. Difficulty also arises in the fact that many indications and symptoms are often ambiguous, and accurate diagnosis can sometimes present a challenge for even trained health experts. Therefore, any model designed to perform such tasks must also be able to robustly handle the inherent ambiguity that comes with the human body, human disease, and medical data. Machine learning models are specifically able to address these needs in comparison to other predictive analytic techniques because they fine-tune their own parameters to best fit the data, allowing them to accurately consider exponentially more features [3].

The selection of the best machine learning model for each classification task remains an area of open research, as the diversity and complexity that exist within and between medical datasets cannot be easily encompassed by one model [4]. Therefore, for a model to be of any real benefit in the healthcare setting, its performance must be evaluated on various medical datasets to identify its strengths and weaknesses. Some studies have shown superior performance of certain models on specific datasets compared to others. For instance, a recent publication reported that on a heart dataset, the naïve Bayes model had the best score across all performance metrics considered. It was followed closely by the k-nearest neighbours model (KNN). [5]. However, in the context of stroke prediction, support vector machine (SVM) and random forest emerged as the best-performing models while KNN and naïve Bayes models were sub-optimal [6]. Similar trends have been reported in other comparative studies that evaluated various models on real-life medical datasets. Evaluating the strengths and weaknesses of various machine learning models on medical data potentially paves the way for more tailored research in various medical domains. The strengths of each model on a

particular dataset can be harnessed to develop systems for medical applications, while limitations of the algorithm can be investigated.

To provide a meaningful contribution, this project seeks to identify, evaluate and compare the classification performance of the best-performing models reported in the literature on open-source medical datasets. This contribution is relevant as various sources in the current literature can give conflicting opinions on the benefits of each method. In addition, trying to empirically account for the variations in datasets and variable specifications for each method between papers, and how this impacts the results is challenging. By choosing the best-performing models according to the current literature and comparing them on an even playing field with open-source methods and datasets, we hope to eliminate some guesswork for those in this field when trying to discern which method is best for any specific application. We also provide a brief summary of the current landscape in machine learning classification for medical datasets, evaluating their suitability for this type of prediction problem.

Various machine learning models could be highly beneficial as diagnostic or prognostic tools in the clinical setting. They are a viable avenue to aid clinicians in some of the more difficult aspects of their work. Therefore, this project seeks to support their development in the medical field by providing a transparent open-source analysis of the most popular methods.

2 Related work

Several studies have compared a multitude of classification algorithms for medical datasets. The goal of reviewing this literature is to provide a summary of the current landscape of machine learning classification for medical datasets, in addition to identifying which of these algorithms tend to perform best so that they can be further evaluated and compared.

A study conducted by Hassan *et al.* [7] evaluated and compared using logistic regression, decision tree, naïve Bayes, KNN, SVM, and random forest for classifying medical data related to heart disease and hepatitis. The study evaluated the performance of these different algorithms using accuracy, precision, recall and F-score. For all of these metrics, it was found that naïve Bayes and KNN had the worst scores, and random forest and SVM had the best scores when applied to the heart disease dataset. When applied to the hepatitis dataset, random forest and logistic regression had the best scores for all metrics. Based on these results, random forest appears to consistently perform best between the two datasets.

In another study conducted by Sidey-Gibbons & Sidey-Gibbons [8], they evaluated and compared regularized general linear model regression, SVM with a radial basis function (RBF) kernel, and single-layer artificial neural network for classifying breast cancer data. The study evaluated the performance of these algorithms using accuracy, sensitivity and specificity. The study found that all three algorithms had similar performance, but SVM achieved the best accuracy of 0.96. A voting ensemble of the three algorithms was also evaluated but provided only marginally increased performance.

Mittal & Gill [9] conducted a comparative analysis of decision tree, naïve Bayes, neural network, SVM and KNN for classifying heart disease, diabetes and hepatitis data. To evaluate and compare the algorithms, the study compared accuracy, Kappa score, execution time and mean absolute error (MAE). Overall, Mittal & Gill found that SVM had the best performance, with the lowest MAE, highest accuracy and highest Kappa score across all datasets. While the execution time was not the shortest, it was still relatively short when compared to the neural network model. The accuracy obtained from applying SVM to the heart disease dataset was 0.84 which is similar to the accuracy of 0.82 obtained by Hassan *et al.* from also applying SVM to heart disease data [7]. However, an accuracy of 0.85 was obtained in this study from applying SVM to the hepatitis dataset whereas Hassan *et al.* obtained 0.72. The difference observed here could be a result of variations in implementation details or hyperparameter choices which were not specified in either study.

Another comparative analysis was conducted by Biswas *et al.* which applied the classifiers for stroke diagnosis prediction [6]. The study evaluated and compared SVM, random forest, KNN, decision tree, naïve Bayes, logistic regression, voting classifier, AdaBoost, gradient boosting, neural network and nearest centroid. The voting classifier was an ensemble of logistic regression and SVM. The study achieved 0.99 accuracy, precision, recall and F-score with both SVM and random forest, making

them the two best-performing classifiers. KNN also performed very well, achieving between 0.98 to 0.99 across all metrics. This contrasts the results obtained by both Hassan *et al.* and Mittal & Gill which showed KNN generally performed quite poorly compared to other classifiers [7], [9].

Many of these studies compared these algorithms on only a limited number of medical datasets. As such, there is still a need to provide a more comprehensive comparison of the performance of these algorithms across various types of medical datasets to evaluate their transferability. It was also observed that variations occurred between studies, even on the same datasets. This can potentially be attributed to the differences in data preprocessing, implementation details and hyperparameter choices, but because the details of these steps were not provided in most of these studies, it can be difficult to identify the exact cause. This makes comparing the performance of different models between different studies challenging. SVM generally performed well across all studies, particularly with an RBF kernel [8]. Random forest, which was evaluated in only two of the studies discussed, was also one of the best-performing classifiers in both studies. We will evaluate and compare these two algorithms on breast cancer diagnostic data, heart disease data, liver disorder data and hepatitis data. This work will contribute towards the development of better diagnostic and prognostic tools in the medical field by providing open-source implementations and analyses.

3 Methods

3.1 Datasets

The datasets used for this project were all obtained from the UC Irvine Machine Learning Repository. The four datasets used were the Cleveland database from *Heart Disease* [10], *Breast Cancer Wisconsin (Diagnostic)* [11], *Liver Disorders* [12], and *Hepatitis* [13]. All four of these datasets can be used for classification with all of them having binary labels except the heart disease dataset which has five classes. These datasets encompass a range of common diseases and present classification challenges in both diagnosis and prognosis.

3.2 Preprocessing

Preprocessing of the data was performed prior to training the models. The heart disease dataset required binning the labels to provide a more robust dataset. Originally, the dataset contained five classes indicating the severity of heart disease from no heart disease (0) to the most severe (5). However, this created a very imbalanced dataset with 160 instances of class 0 and only 13 instances of class 5. Instead, this was changed to a binary problem by binning the classes greater than 0 into a single class. Studies that have used this dataset in the past have also taken this approach [10]. The liver disorders dataset also required binning the labels to be used for binary classification. Examples with "drinks" > 3 were combined into a single class and examples with "drinks" ≤ 3 were combined into another class. This was the same approach used in the study conducted by Turney [14], and was suggested by the providers of the dataset [12].

The heart disease and hepatitis datasets contained some missing values. For the heart disease dataset, this was handled by simply removing the examples which contained missing values. Only six examples contained features with missing values so this was not a significant reduction. However, the hepatitis dataset contained a greater proportion of examples with missing data. Instead of removing these examples, the missing data was imputed with the mean of the corresponding feature. There was no need to encode any features because all the categorical features in all four datasets were already encoded to numerical values.

$$x_{ij} = \frac{x_{ij} - \text{median}(x_j)}{\text{interquartile range of } x_j} \tag{1}$$

Each dataset was then randomly split into corresponding training and testing sets with a 4:1 ratio. The features of all four datasets were then standardized using a robust scalar. Robust scaling was used instead of standard scaling due to the presence of outliers in some of the features of the datasets. The specific equation for scaling the values is expressed in equation (1). Care was taken to ensure the scaling of the testing sets was performed with the median and interquartile range of the training sets.

3.3 Training

Two models were trained based on SVM with an RBF kernel and random forest respectively for each of the four datasets. For each model, we conducted a grid search with five-fold cross-validation to optimize the hyperparameters (`C = [0.01, 0.1, 1, 10, 100]` for SVM and `n_estimators = [10, 50, 100, 150, 200]` for random forest). The training was implemented with the scikit-learn library in Python. Access to the repository for the source code can be found in Appendix A.

3.4 Evaluation

The metrics used in evaluating the performance of the models were accuracy, precision, recall, and the F1 score. These metrics provide insight into the proportion of instances of the medical condition that are correctly classified. Accuracy measures the overall correctness of the model's predictions, providing insight into the proportion of correctly classified examples among all test examples. Precision emphasizes the accuracy of positive predictions, thus reflecting the reliability of the model in correctly identifying instances of a specific medical condition. It is calculated as the ratio of true positives to the sum of true positives and false positives. Recall quantifies the model's ability to capture all positive instances by measuring the ratio of true positives to the sum of true positives and false negatives. In medical classification, precision and recall as independent metrics do not provide a full evaluation of the model's performance and so must be examined together. In other words, a model with a low false positive rate has a high precision score, and a model with a low false negative rate has a high recall score. The F1 score provides a combined measure of these two metrics by considering both false positives and false negatives. Equation (2) illustrates how these metrics were calculated.

$$\begin{aligned} \text{Accuracy} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \\ \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}} \\ \text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{FN}} \\ \text{F1 Score} &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned} \tag{2}$$

Where,
TP =Number of true positives,
TN=Number of true negatives,
FP =Number of false positives,
FN=Number of false negatives

4 Results

The bar graphs in Figure 1 compare the various metrics obtained by both algorithms on the different datasets. From observation, random forest obtained a slightly better accuracy on most datasets with a maximum difference of 0.066. Both algorithms obtained an accuracy of over 0.8 on all datasets except the liver disorders dataset. The accuracies are displayed in Table 1. Random forest generally achieved a higher precision and F1-score compared to SVM. Additionally, both algorithms achieved almost perfect accuracy on the breast cancer dataset. Looking at Figure 2, we can observe that SVM made just two errors and random forest made just one error. The confusion matrices for the other datasets are shown in Appendix B.

Table 1: Accuracy of SVM and random forest achieved for each dataset

Dataset	SVM accuracy	Random forest accuracy
Heart Disease	0.817	0.883
Breast Cancer Wisconsin	0.982	0.991
Liver Disorders	0.565	0.522
Hepatitis	0.839	0.903

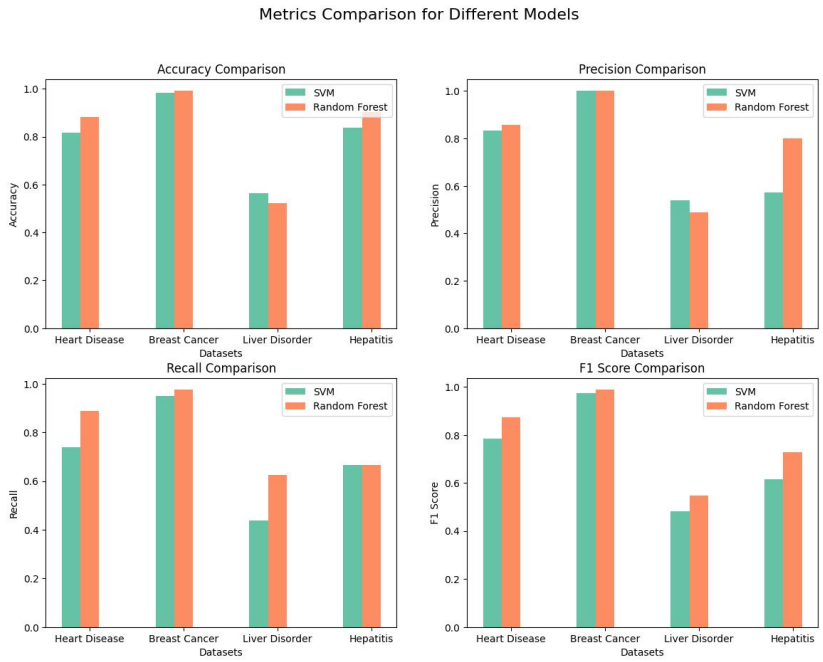

Figure 1: Bar graphs comparing accuracy, precision, recall and F1 score between the SVM and random forest classifiers across all datasets

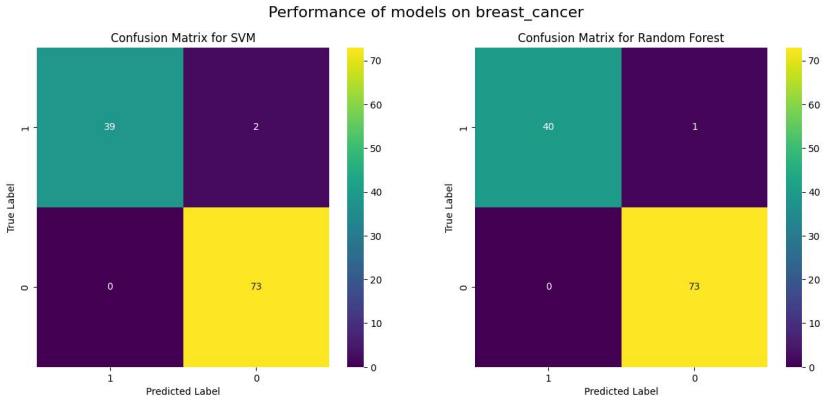

Figure 2: Confusion matrix for SVM and random forest applied on the *Breast Cancer Wisconsin* dataset

5 Discussion

In accordance with findings reported in existing literature pertaining to the performance of SVM and random forest, general patterns emerge across multiple datasets, accompanied by variations in performance when comparing the model's predictive strength within a specific dataset to its performance in another. Nevertheless, in our comprehensive assessment, random forest consistently demonstrated superior performance relative to SVM with RBF across most considered evaluation metrics and datasets. This implies that within the context of the investigated medical datasets, random forest exhibited a higher overall predictive capacity. When contrasting performance across datasets, both models achieved better accuracy scores on the breast cancer and hepatitis datasets compared to

the heart disease classification task, with random forest achieving higher accuracy, precision, recall, and F1 scores. Additionally, the accuracy achieved by SVM on the heart disease dataset closely aligns with reported results by Hassan *et al.* [7] and Mittal & Gill [9]. Yet, the accuracy score obtained for random forest in the heart dataset surpassed the score reported by Hassan *et al.* [7] and the SVM score obtained by Mittal & Gill [9].

Analogous trends are observed in the comparison of our accuracy scores against the accuracy scores obtained by Hassan *et al.* [7] and Mittal & Gill [9] for the hepatitis data. Specifically, while these studies reported a maximum accuracy of 0.72 and 0.85 respectively for this dataset, we obtained a superior accuracy of 0.90 with random forest. In the breast lesion classification task, Sidey-Gibbons & Sidey-Gibbons [8] reported their best accuracy of 0.96 with the SVM model. However, in this current study, we obtained a slightly higher accuracy of 0.98 with SVM and 0.99 with random forest, accompanied by notably high F1 scores of 0.98 and 0.99 with SVM and random forest respectively. Although the comparative studies did not provide details of the techniques employed in training their models, the differences in metric scores can potentially be attributed to the choice of hyperparameters.

In the classification of liver disorders, both models encountered challenges in accurately predicting liver disorders, and this is evident from the lower scores obtained. SVM with RBF achieved marginally better accuracy than random forest in this specific dataset; however, random forest exhibited an F1 score surpassing that of SVM. Nonetheless, both models consistently maintained metric values lower than 0.6. This observation could be attributed to inherent challenges associated with mapping features to labels in this specific dataset. The dataset is structured to predict the level of alcohol intake based on physiological markers, yet it may lack an inherent correlation between these markers and the predicted outcome. Thus, to train models to classify liver disorders, using a dataset that maps features to more relevant labels, that is, if a person may be diagnosed with a liver disorder or not, may potentially improve performance.

In the clinical diagnostic context, particularly for high-risk diseases, having a high recall is more important than precision. Based on this, random forest is likely to be more useful as it achieved equal or higher recall than SVM with RBF for the heart disease, breast cancer, and liver disorder datasets. These datasets all present a diagnostic problem. With that said, random forest likely would not be useful for liver disorder data as the recall is still relatively low. To further motivate the benefit of random forest, SVM with RBF may lack the interpretability afforded by random forest. The latter, being an ensemble of decision trees, provides feature importance rankings that could assist clinicians in understanding the significance of distinct features in the classification process. Additionally, while SVM with RBF exhibited competitive performance on select datasets, its transferability across varied medical datasets may be constrained.

6 Conclusion and future work

In this work, we performed an open-source implementation and analysis of two popular machine learning methods applied to four freely available medical datasets for predicting disease diagnosis and prognosis. These methods were SVM with an RBF kernel and random forest. This endeavour was undertaken with the intent of providing a transparent comparison of these methods on a level playing field, allowing researchers to easily assess their performance under varying applications. This allows them to avoid the trial-and-error process when selecting a method for their application. We concluded that random forest generally performed best on most metrics across all datasets, often obtaining comparable or better accuracy values to similar implementations in current literature. However, both methods seemed to encounter challenges with the liver disorder dataset, highlighting the importance of not assuming a method's transferability across varied medical datasets.

Future directions of this research can include performing similar analyses on a greater variety of medical datasets and larger datasets (regarding both examples and features). Furthermore, expanding towards multi-class implementations of these methods can be explored. Another possible future direction is investigating how the preset features of the datasets are chosen and instead potentially using neural network-based methods to allow the model itself to learn the best features. In summary, our open-source analysis supports and fosters the development of robust machine learning methods that have the potential to be highly beneficial diagnostic tools in clinical settings.

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Appendix

A Source code

The source code for this project can be accessed at `https://github.com/ryanyeung1/CPSC540-MedicalDataClassifier.git`

B Confusion matrices

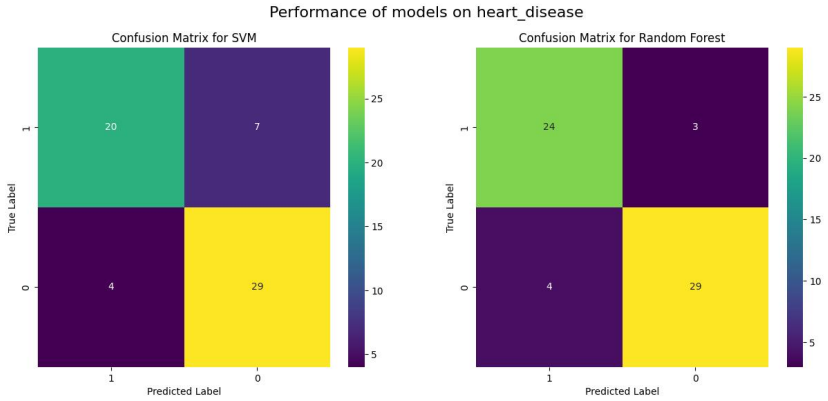

Figure 3: Confusion matrix for SVM and random forest applied on the *Heart Disease* dataset

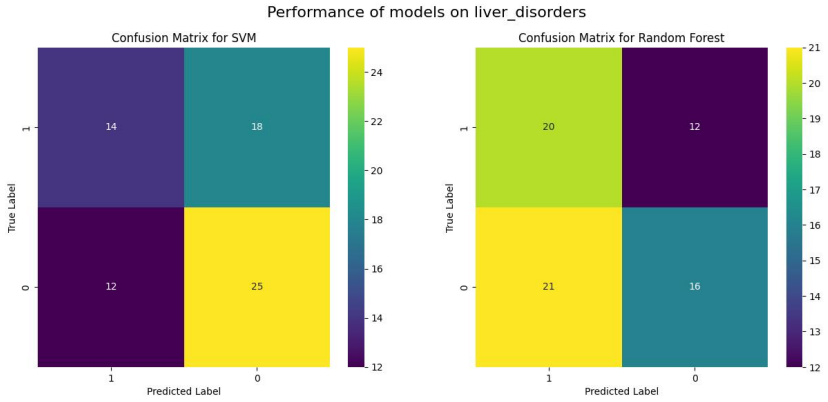

Figure 4: Confusion matrix for SVM and random forest applied on the *Liver Disorders* dataset

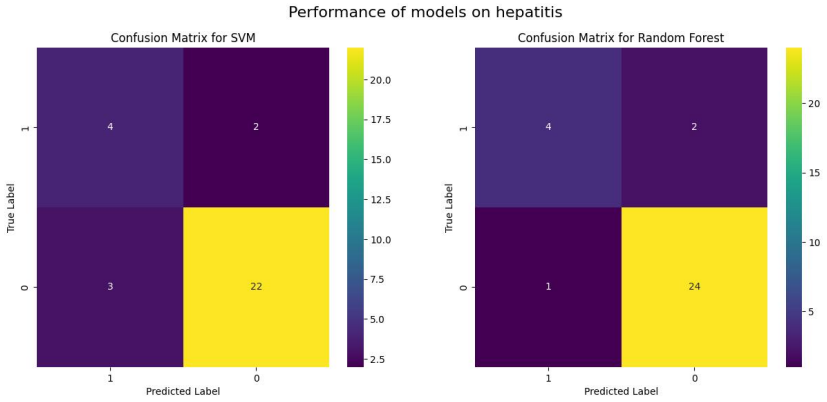

Figure 5: Confusion matrix for SVM and random forest applied on the *Hepatitis* dataset